① p-series $\sum \frac{1}{n^p}$ conv. if $p>1$.

② Geo. series $\sum ar^n$ conv. if $-1<r<1$.

Lecture 19: Comparison Tests (10.3)

II

$\sum \frac{1}{3^n-2}$

$\frac{1}{3^n-2} > \frac{1}{3^n}$

ex. Compare the two series:

$$\sum_{n=1}^{\infty} \frac{1}{3^n+2} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{3^n}.$$

conv.

: Geo series $r = \frac{1}{3}$

Since
$$\frac{1}{3^n+2} < \frac{1}{3^n} \text{ for } n \geq 1,$$

$$\sum_{n=1}^{\infty} \frac{1}{3^n+2} < \sum_{n=1}^{\infty} \frac{1}{3^n} \text{ for } n \geq 1.$$

conv.

Since the bigger series converges (geometric series), the smaller series must converges, too.

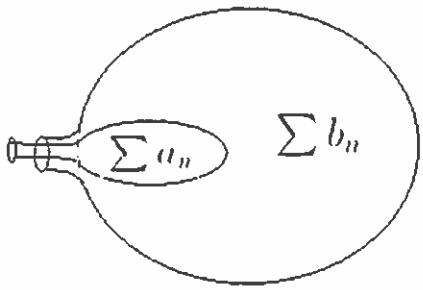
Direct (or Basic) Comparison Test:

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms, and if $b_n \geq a_n$ for all n, then:

1. If $\sum_{n=1}^{\infty} b_n$ converges, then so does $\sum_{n=1}^{\infty} a_n$.
2. If $\sum_{n=1}^{\infty} a_n$ diverges, then so does $\sum_{n=1}^{\infty} b_n$.

When you compare 2 series, there are 4 possibilities. The first two are:

1 If the bigger series converges, so does the smaller series and, if 2 the smaller series diverges, so does the bigger series.



balloon analogy

ex. Determine the convergence of the series:

(1) $\displaystyle\sum_{n=4}^{\infty} \frac{1}{\sqrt{n}-1}$ $\qquad \sum \frac{1}{\sqrt{n}}$

$\frac{1}{\sqrt{n}-1} > \frac{1}{\sqrt{n}}$ and $\sum \frac{1}{\sqrt{n}}$ $(p=\frac{1}{2})$ div.

$\Rightarrow \sum \frac{1}{\sqrt{n}-1}$ div. by the comp. test.

(2) $\displaystyle\sum_{n=10}^{\infty} \frac{1}{n^2+5}$ $\qquad \sum \frac{1}{n^2}$

$\frac{1}{n^2+5} < \frac{1}{n^2}$ and $\sum \frac{1}{n^2}$ $(p=2)$ conv.

$\Rightarrow \sum \frac{1}{n^2+5}$ conv. by comp. test.

The other 2 possibilities are not so conclusive: If the smaller series converges, or if the bigger series diverges, we get absolutely no information about the other series.

(3) $\displaystyle\sum_{n=4}^{\infty} \frac{1}{\sqrt{n}+1}$? $\qquad \sum \frac{1}{\sqrt{n}}$

$\frac{1}{\sqrt{n}+1} < \frac{1}{\sqrt{n}}$ and $\sum \frac{1}{\sqrt{n}}$ $(p=\frac{1}{2})$ div.

The comp. test does not work.

That leads to our next comparison test:

The Limit Comparison Test Suppose that $\sum a_n$ and $\sum b_n$ are both positive term series, and

finite positive #.

$\frac{a_n}{b_n} \to 2$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \boxed{c,} \quad 0 < c < \infty$$

then either both series converge or both series diverge.

$\sum b_n \approx c \sum a_n$

Instead of setting up an **inequality** to **directly** compare the sizes of the terms of the two series, here we compare the **rate of growth** of the terms of the series. And it doesn't matter whether we take $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ or $\lim_{n\to\infty} \frac{b_n}{a_n} = \frac{1}{c}$, the results are the same.

$\frac{a_n}{b_n} \to \infty \qquad a_n >> b_n$

(**Note also:** If $c = \infty$ and $\sum a_n$ converges, then $\sum b_n$ converges. (or if $\sum b_n$ div. $\Rightarrow \sum a_n$ div.)

If $c = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges)

$\frac{a_n}{b_n} \to 0 \quad , \quad b_n >> a_n$

Determine the convergence or divergence of the series, which test do you use?

(1) $\sum_{n=4}^{\infty} \frac{1}{\sqrt{n}+1}$ div. by LCT. $\sum \frac{1}{\sqrt{n}}$ $(p=\frac{1}{2})$ div.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{\sqrt{n}+1}}{\frac{1}{\sqrt{n}}} = \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n}+1} = 1$$

$x \to 0.$

$\sin x \approx x$

(2) $\sum_{n=2}^{\infty} \sin\left(\frac{1}{n}\right)$ div. by LCT. $\sum \frac{1}{n}$ $(p=1)$ div.

$$\lim_{n\to\infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} \underset{\theta=\frac{1}{n}}{=} \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$$

(2)' $\sum_{n=2}^{\infty} \sin^2\left(\frac{1}{n}\right)$ conv. by LCT. $\sum \frac{1}{n^2}$ $(p=2)$ conv.

$$\lim_{n\to\infty} \frac{\sin^2\left(\frac{1}{n}\right)}{\frac{1}{n^2}} = \lim_{\theta\to 0} \frac{\sin^2\theta}{\theta^2} = \left(\frac{\sin\theta}{\theta}\right)^2 \longrightarrow 1$$

(3) $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n^2-1}}{n^3+2n^2+5}$: conv. by LCT.

$$\sum_{n=1}^{\infty} \frac{\sqrt{n^2}}{n^3} = \sum_{n=1}^{\infty} \frac{n}{n^3} = \sum \frac{1}{n^2}$$

conv. $p=2$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\sqrt{n^2-1}}{n^3+2n^2+5} \cdot \frac{n^2}{1} = \lim_{n\to\infty} \frac{\sqrt{n^2}\cdot n^2}{n^3} = 1$$

NYTI:

1. Ralph hopes to investigate the convergence of $\displaystyle\sum_{n=1}^{\infty} \frac{e^{-n}}{n}$ by comparing it with $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$.

Is Ralph on the right track? Explain.

Determine the convergence of the series and which test do you use.

$\lim_{n\to\infty} (-1)^n$ DNE $\Rightarrow$ Do not use LCT.

2. $\displaystyle\sum_{n=1}^{\infty} \frac{2+(-1)^n}{n\sqrt{n}}$

conv. by comp. test.

$$\frac{2-1}{n\sqrt{n}} \le \frac{2+(-1)^n}{n\sqrt{n}} \le \frac{2+1}{n\sqrt{n}}$$

$\sum \frac{1}{n\sqrt{n}}$ $(p=3/2)$ conv.

$$\frac{1}{n\sqrt{n}} \le \boxed{\frac{2+(-1)^n}{n\sqrt{n}} \le \frac{3}{n\sqrt{n}}}$$

3. $\displaystyle\sum_{n=1}^{\infty} \frac{2n^2+7n}{3^n(n^2+5n-1)}$

$$\sum \frac{2n^2}{3^n n^2} = \sum \frac{2}{3^n}$$

$r = \frac{1}{3}$

LCT.

4. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^5+100}}$

5. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)!}$ Conv. by comp. test.

$$\frac{1}{(n+1)!} \le \frac{1}{n(n+1)}$$

$$a_n = \frac{1}{(n+1)!} = \frac{1}{1\cdot 2\cdot 3\cdots n(n+1)} = \left(\frac{1}{1}\right)\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)\cdots\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right)$$

$$\le 1\cdot 1\cdot \cdots 1\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right)$$

$\sum \frac{1}{n(n+1)} \sim \sum \frac{1}{n^2}$ conv.

$\sum_{n=1}^{\infty} \frac{1}{n+1}$ $(p=1)$ div.

6. $\displaystyle\sum_{n=8}^{\infty} \frac{42\sqrt{n}}{5n^2 - 9n + 1}$

7. $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$

8. $\displaystyle\sum_{n=1}^{\infty} \frac{1 + 2^n}{1 + 3^n}$ conv. by LCT.

$\sum \frac{2^n}{3^n}$ conv. $\left(r = \frac{2}{3}\right)$

$\lim \sin n$ DNE. $\Rightarrow$ no LCT.

9. $\displaystyle\sum_{n=1}^{\infty} \frac{1 + \sin n}{10^n}$ Conv. by comp. test.

$$0 = \frac{1-1}{10^n} \le \underbrace{\frac{1+\sin n}{10^n} \le \frac{1+1}{10^n} = \frac{2}{10^n}}$$

$\sum \frac{2}{10^n}$ conv. $r = \frac{1}{10}$

(3) $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n^2-1}}{n^3+2n^2+5}$

NYTI:

1. Ralph hopes to investigate the convergence of $\displaystyle\sum_{n=1}^{\infty} \frac{e^{-n}}{n}$ by comparing it with $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$.

Is Ralph on the right track? Explain.

No!

$\frac{e^{-n}}{n} < \frac{1}{n}$ and $\Sigma \frac{1}{n}$ div.

Therefore, the comparison test is inconclusive.

Determine the convergence of the series and which test do you use.

2. $\displaystyle\sum_{n=1}^{\infty} \frac{2+(-1)^n}{n\sqrt{n}}$ conv. by the comparison test.

$$\frac{2+(-1)^n}{n\sqrt{n}} \le \frac{2+1}{n\sqrt{n}} = \frac{3}{n^{3/2}} \text{ and } \sum \frac{3}{n^{3/2}} \text{ conv. } (p=3/2)$$

3. $\displaystyle\sum_{n=1}^{\infty} \frac{2n^2+7n}{3^n(n^2+5n-1)}$ Conv. by LCT.

$$b_n = \frac{2n^2}{3^n \cdot n^2} \qquad \lim_{n\to\infty} \frac{2n^2+7n}{\cancel{3^n}(n^2+5n-1)} \cdot \frac{\cancel{3^n} \cdot n^2}{2n^2} = 1 \text{ and } \sum \frac{2n^2}{3^n n^2}$$

$$= \sum \frac{2}{3^n} \text{ conv. } \left(r = \frac{1}{3}\right)$$

4. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^5+100}}$ conv. by the comparison test.

$$\frac{1}{\sqrt{n^5+100}} \le \frac{1}{\sqrt{n^5}} \text{ and } \sum \frac{1}{n^{5/2}} \text{ conv. } (p=5/2)$$

5. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)!}$ Conv. by the comparison test.

$$a_n = \frac{1}{(n+1)!} = \frac{1}{1\cdot 2\cdot 3 \cdots n\cdot(n+1)} = \underset{\le 1}{\left(\frac{1}{1}\right)} \underset{\le 1}{\left(\frac{1}{2}\right)} \underset{\le 1}{\left(\frac{1}{3}\right)} \cdots \overbrace{\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right)}^{\text{keep two}}$$

$$\le 1 \cdot 1 \cdot \cdots 1 \cdot \frac{1}{n} \cdot \frac{1}{n+1}$$

$$a_n \le \frac{1}{n(n+1)} \le \frac{1}{n^2}$$

and $\sum \frac{1}{n^2}$ conv. $(p=2)$

6. $\displaystyle\sum_{n=8}^{\infty} \frac{42\sqrt{n}}{5n^2 - 9n + 1}$ Conv. by LCT.

$b_n = \dfrac{42\sqrt{n}}{5n^2}$, $\displaystyle\lim_{n\to\infty} \frac{\cancel{42\sqrt{n}}}{5n^2-9n+1} \cdot \frac{5n^2}{\cancel{42\sqrt{n}}} = 1$ and $\sum \dfrac{42\sqrt{n}}{5n^2}$

$= \dfrac{42}{5} \sum \dfrac{1}{n^{3/2}}$ conv.

$(p = 3/2)$

7. $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$ Conv. by the comparison test.

$$a_n = \frac{n!}{n^n} = \frac{1 \cdot 2 \cdot 3 \cdot \cdots \cdot n}{n \cdot n \cdot n \cdots n} = \overbrace{\frac{1}{n} \cdot \frac{2}{n}}^{\text{keep two}} \cdot \underbrace{\frac{3}{n} \cdots \frac{n}{n}}_{1 \cdots 1} \leq \frac{1}{n} \cdot \frac{2}{n} \cdot 1 \cdots 1 = \frac{2}{n^2}$$

$a_n \leq \dfrac{2}{n^2}$ and $\sum \dfrac{2}{n^2}$ conv.

$(p=2)$

8. $\displaystyle\sum_{n=1}^{\infty} \frac{1+2^n}{1+3^n}$ conv. by LCT.

$b_n = \dfrac{2^n}{3^n}$, $\displaystyle\lim_{n\to\infty} \frac{1+2^n}{1+3^n} \cdot \frac{3^n}{2^n} = 1$ and $\sum \dfrac{2^n}{3^n} = \sum \left(\dfrac{2}{3}\right)^n$ conv.

$(r = \frac{2}{3})$

9. $\displaystyle\sum_{n=1}^{\infty} \frac{1+\sin n}{10^n}$ Conv. by the comparison test.

$\dfrac{1+\sin n}{10^n} \leq \dfrac{1+1}{10^n} = \dfrac{2}{10^n}$ and $\sum \dfrac{2}{10^n}$ conv. $(r = \frac{1}{10})$